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Indian Institute of Technology Dharwad

CS101C: Introduction to Programming (Using C)

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CSE Department

Indian Institute of Technology, Dharwad

Week 4

Today's Class (24/8/2026)

Repetitions

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"Do I really have to write this 10 times?"



Suppose we want to print the table of **5**

```
1    printf("%d\n", 5 * 1);      → 5
2    printf("%d\n", 5 * 2);      → 10
3    printf("%d\n", 5 * 3);      → 15
4    printf("%d\n", 5 * 4);      → 20
5    printf("%d\n", 5 * 5);      → 25
6    printf("%d\n", 5 * 6);      → 30
7    printf("%d\n", 5 * 7);      → 35
8    printf("%d\n", 5 * 8);      → 40
9    printf("%d\n", 5 * 9);      → 45
10   printf("%d\n", 5 * 10);     → 50
```







$5 \times 1 \rightarrow 5$, $5 \times 2 \rightarrow 10$, $5 \times 3 \rightarrow 15$, $5 \times 4 \rightarrow 20$, $\cdots$, $5 \times 10 \rightarrow 50$

Only the **multiplier** is changing!





This is the motivation for

LOOP



A loop lets the computer **repeat** a block of code again and again so we **don't have to write it many times!** 😊

if Vs while

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Connecting to What You Already Know!

You already know **if**!

Now let's see how **while** is just an extension of **if**!

You already know:

```
if (condition)
{
    statement;
}
```

Meaning

Check the condition **ONCE**.
If it's **true** → execute the statement.

Now learn:

```
while (condition)
{
    statement;
}
```

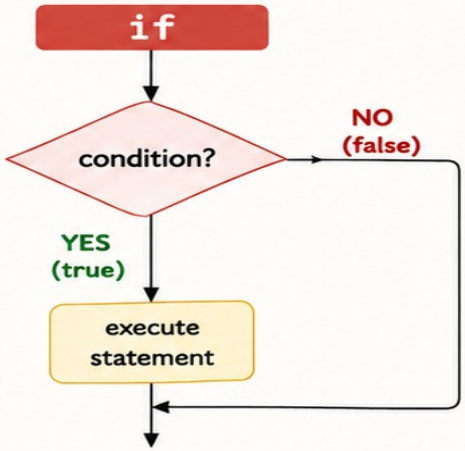
Meaning

Keep checking the condition **AGAIN** and **AGAIN**.
While it's **true** → execute the statement.

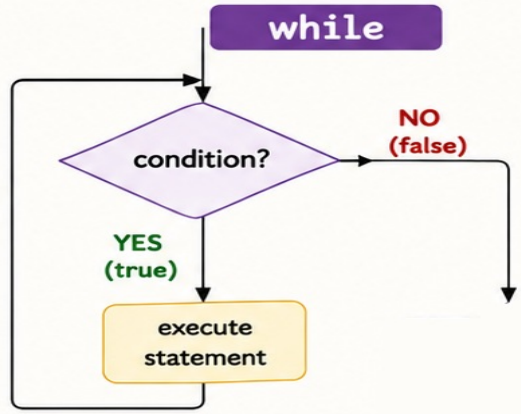
if checks once...



Decide **ONCE**. Then move on.

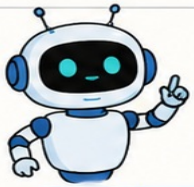


VS



while keeps checking...





if = decide once
Do it or skip it.

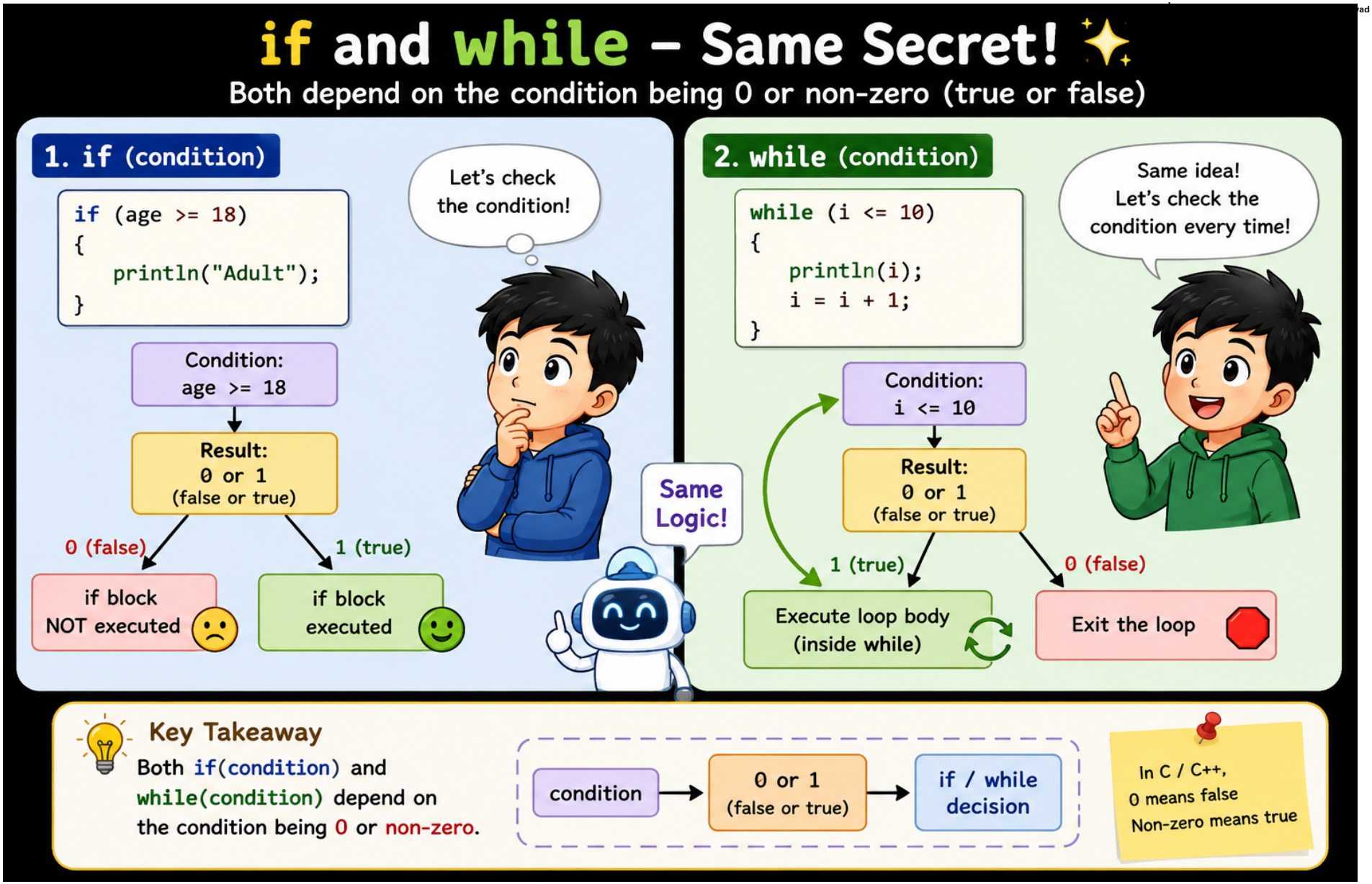
while = repeat while **true**
Do it, again and again as long as it's **true**.

while is just **if** on replay!

Think of **while** as: "If the answer is **YES**, do it and come back to ask again!"

if Vs while

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Your First `while` Loop

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How many times will `printf` execute?

Don't run it yet. **Predict first!**

Program

```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
    i++;
}
```

Question: How many times does `printf` execute?

1? 3? 5? 10?

Let's Trace the Loop

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Watch the condition change!

Every iteration follows the same pattern

Check

$\Downarrow$

YES?

$\Downarrow$

Execute

$\Downarrow$

Update $i++$

$\Downarrow$

Check again

Trace

i	$i \leq 5$	Action
1	YES	Print 1
2	YES	Print 2
3	YES	Print 3
4	YES	Print 4
5	YES	Print 5
6	NO	STOP

1	2	3	4	5

When Does the Loop Stop?

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```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
    i++;
}
```

W

hat eventually makes

$$i \leq 5$$

false?

When Does the Loop Stop?

```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
    i++;
}
```

W

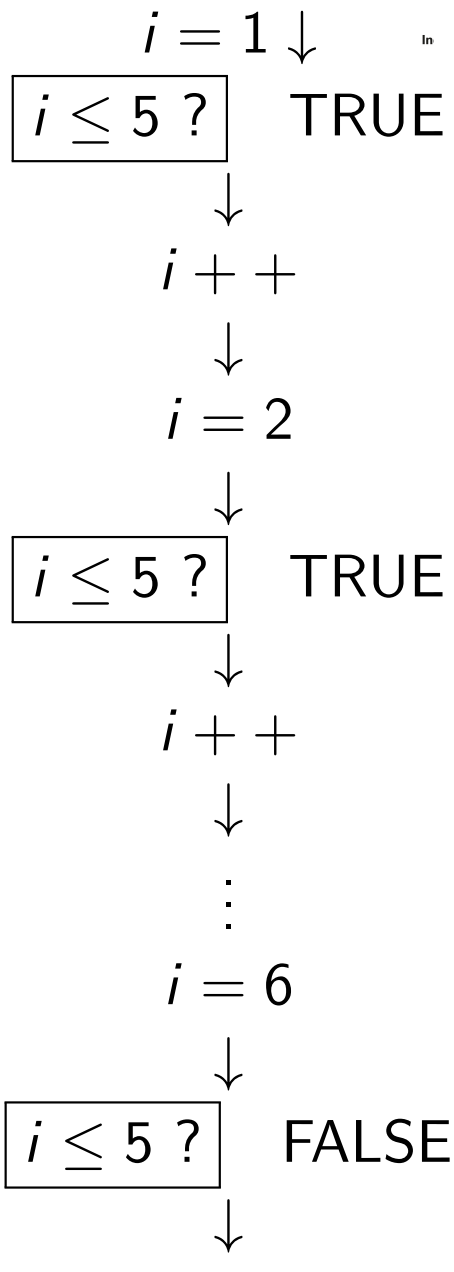
hat eventually makes

$$i \leq 5$$

false?

Answer

i++



The Trap!!

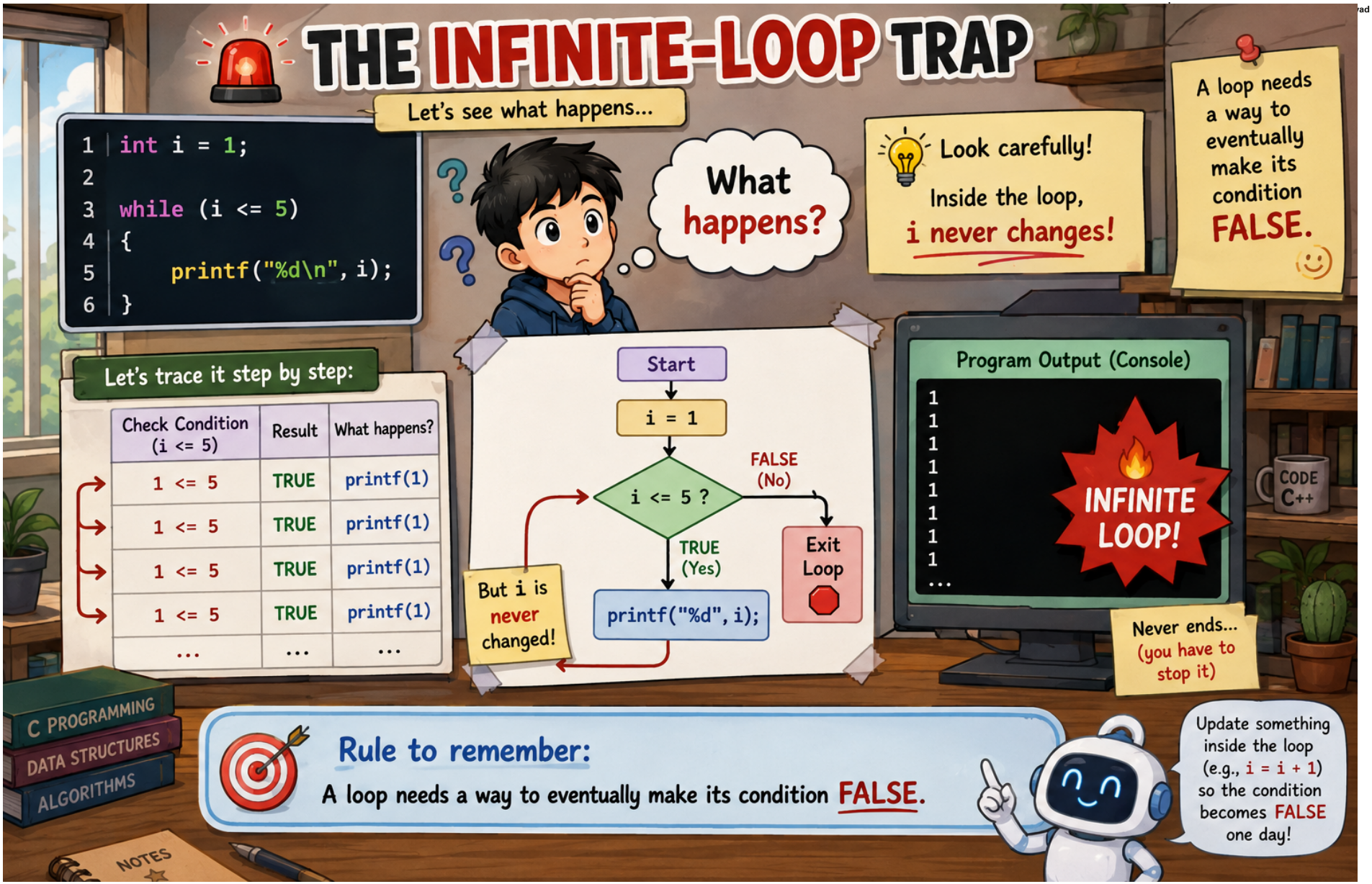

```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
}
```

Output?

The Infinite Loop Trap!!

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A Real Example: Multiplication Table


```
int digit;
int i = 1;

printf("Enter a digit\n");
scanf("%d", &digit);

while(i <= 10)
{
    printf("%d times %d is %d\n",
           digit, i, digit*i);

    i += 1;
}
```

A Real Example: Multiplication Table

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```
int digit;
int i = 1;

printf("Enter a digit\n");
scanf("%d", &digit);

while(i <= 10)
{
    printf("%d times %d is %d\n",
           digit, i, digit*i);

    i += 1;
}
```

Input

```
Enter a digit

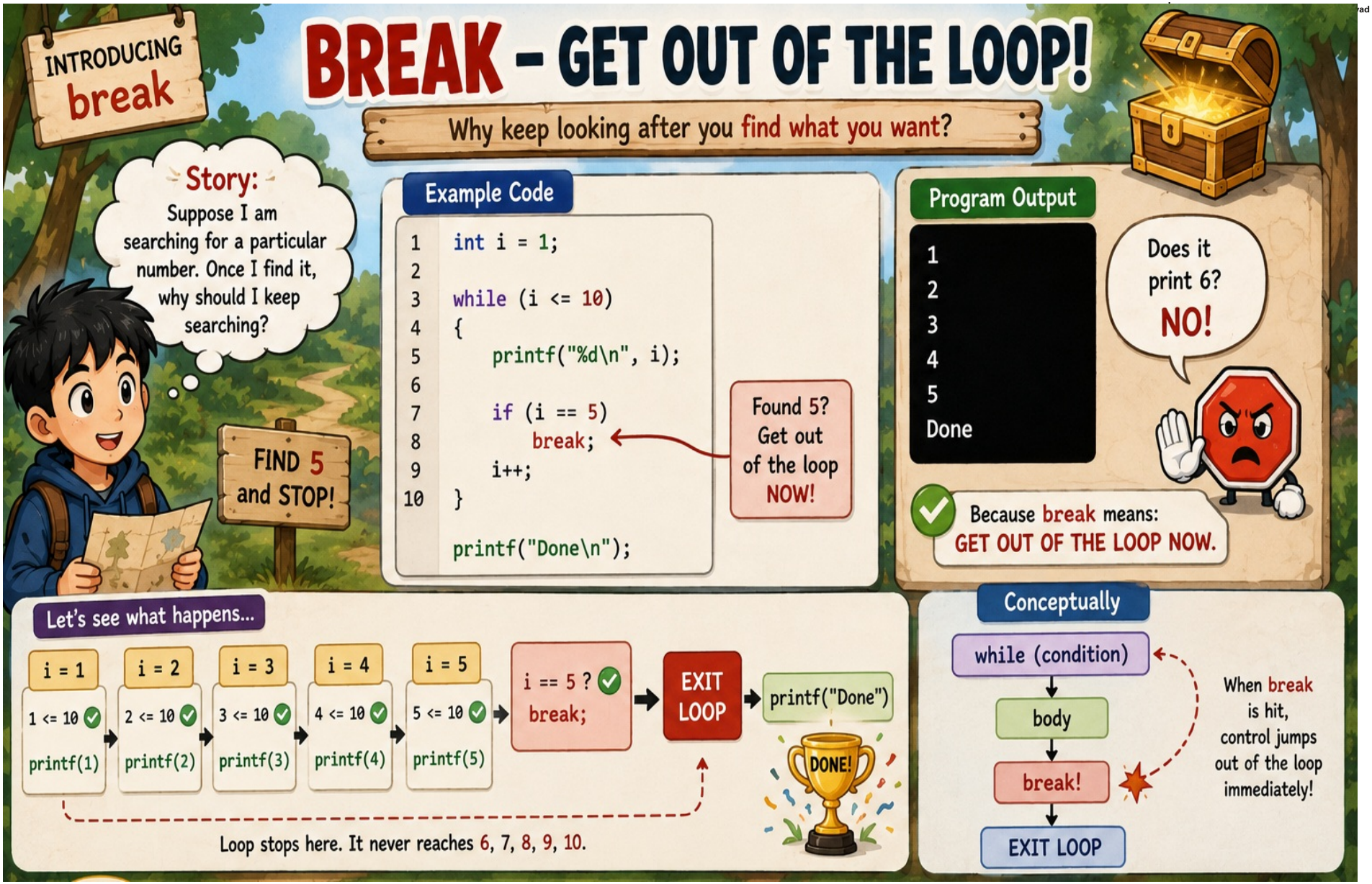
5
```

Output

```
5 times 1 is 5
5 times 2 is 10
5 times 3 is 15
5 times 4 is 20
5 times 5 is 25
5 times 6 is 30
5 times 7 is 35
5 times 8 is 40
5 times 9 is 45
5 times 10 is 50
```

Break Statement

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Continue Statement

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NOT STOPPING... JUST SKIPPING!

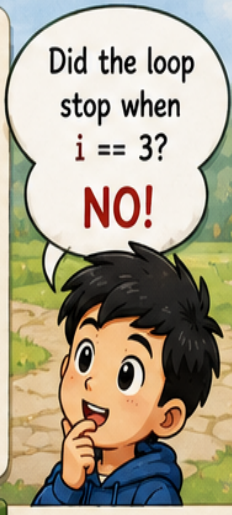
CONTINUE – SKIP & MOVE ON!

NEXT ITERATION

Skip the rest of THIS iteration and go to the next one.

Example Code

```
1   int i = 0;
2
3   while (i < 5)
4   {
5       i++;
6
7       if (i == 3)
8           continue;   // Skip the rest
9
10      printf("%d\n", i);
11  }
```



Program Output

```
1
2
4
5
```

Notice: 3 is missing!
We skipped printing 3, but the loop didn't stop.

What does continue do?

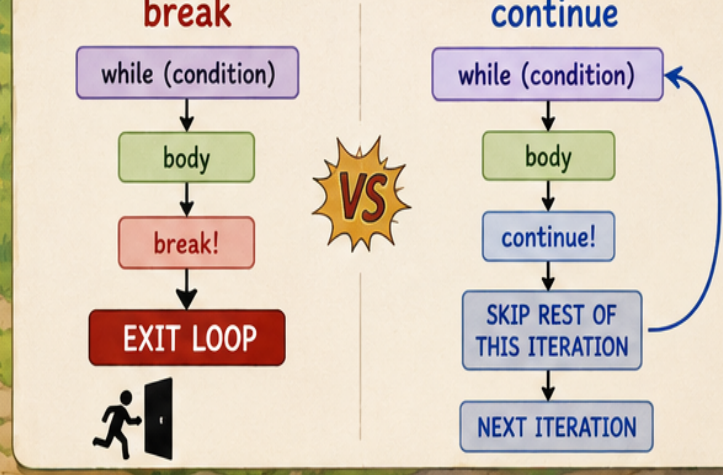
continue means:
"Skip the rest of THIS iteration and go to the next iteration."



Let's trace it step by step...

Iteration	Start: i	After i++	Is i == 3?	Action	Output	Where does it go next?
1	0	1	No	print 1	1	Next iteration
2	1	2	No	print 2	1 2	Next iteration
3	2	3	Yes	continue; (skip print)	1 2	Skip rest → Next iteration
4	3	4	No	print 4	1 2 4	Next iteration
5	4	5	No	print 5	1 2 4 5	Next iteration
6	5	(loop condition fails)	—	Exit loop	1 2 4 5	Loop ends

break vs continue



Loop Detective

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The Mystery Code

```
int i = 1;

while (i <= 10)
{
    if (i % 2 == 0)
    {
        printf("%d\n", i);
    }

    i++;
}
```

Trace the loop carefully and become a **Loop Detective!**

Questions

1. What will be printed?
2. How many times does the `while` body execute?
3. How many times is the `printf` executed?
4. How many times is the `while` condition checked?

Loop Detective

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The Mystery Code

```
int i = 1;

while (i <= 10)
{
    if (i % 2 == 0)
    {
        printf("%d\n", i);
    }

    i++;
}
```

Trace the loop carefully and become a **Loop Detective!**

Questions

1. What will be printed?
2. How many times does the `while` body execute?
3. How many times is the `printf` executed?
4. How many times is the `while` condition checked?

Output:

2, 4, 6, 8, 10

Counts:

- Loop body → **10 times**
- `printf` → **5 times**
- Condition → **11 times**

Challenge: Palindrome Number



Problem

Write a C program using a **while loop** to check whether a number is a palindrome.

Example 1

```
Enter a number:  1221
```

$$1221 = 1221$$

```
1221 is a palindrome
```

Example 2

```
Enter a number:  1234
```

$$1234 \neq 4321$$

```
1234 is not a palindrome
```

Think!

A palindrome reads the same **forward and backward**.

$$1221 \quad \longleftrightarrow \quad 1221$$

Challenge Rule

Do NOT convert the number into a string!

Solution: Palindrome Number

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```
#include <stdio.h>
int main()
{   int n, original, reverse = 0;
    int digit;
    printf("Enter a number: ");
    scanf("%d", &n);
    original = n;
    while (n != 0)
    {
        digit = n % 10;
        reverse = reverse * 10 + digit;
        n = n / 10;
    }
    if (original == reverse)
        printf("%d is a palindrome\n", original);
    else
        printf("%d is not a palindrome\n", original);
    return 0;}
```

Challenge: Can You Print This Diamond?

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Write a C program using while loop to print:

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

Hint:

```
Think about spaces + stars + rows
```

Rule

Use **nested loops**. No hard-coded rows!

Today's Class (28/8/2026)

Recap: While Loop!!

Example: while

```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
    i++;
}
```

"Before printing even the first number, what does the computer do?"

Recap: While Loop!!

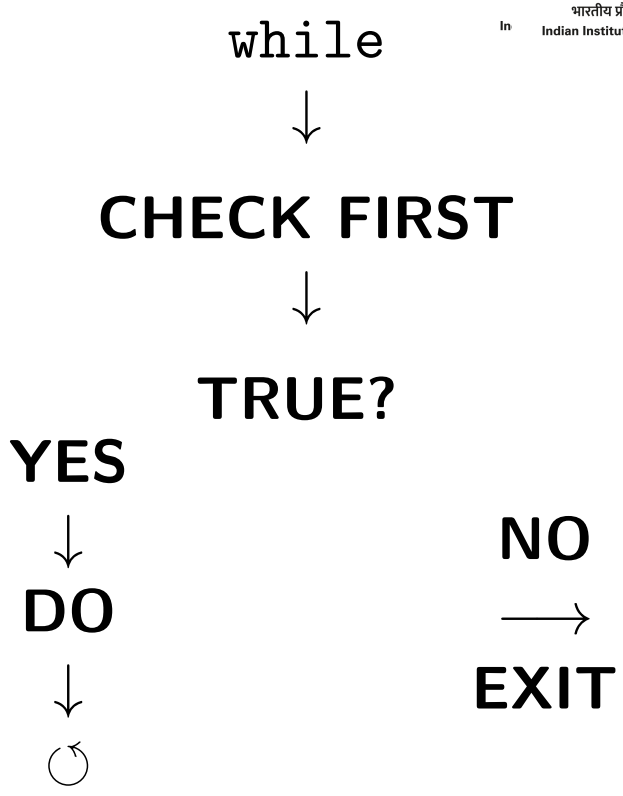
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Example: while

```
int i = 1;

while (i <= 5)
{
    printf("%d\n", i);
    i++;
}
```

"Before printing even the first number, what does the computer do?"



What if I want the computer to **DO something first** and **only THEN** check the condition?

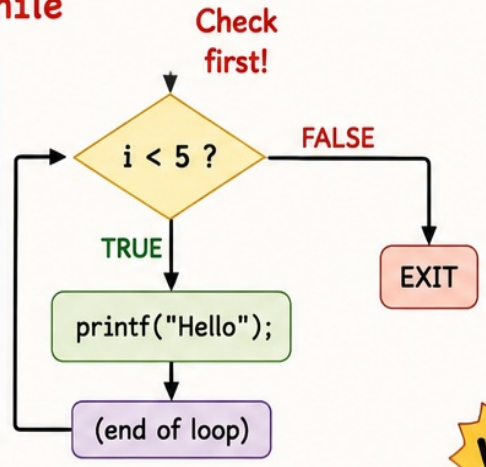
While Vs Do While

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1 Using while

```
int i = 10;

while (i < 5)
{
    printf("Hello\n");
}
```





Let's trace:
i = 10
↓
i < 5 ?
↓
10 < 5 → FALSE
↓
EXIT

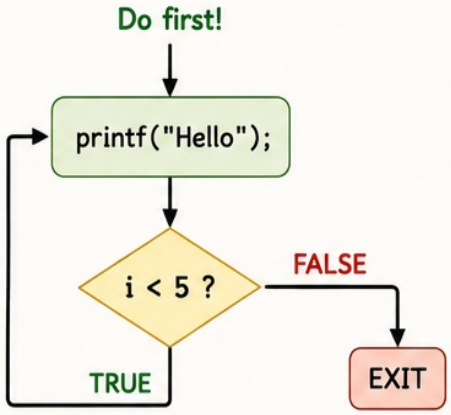
Output:
nothing

VS

2 Change only while to do-while

```
int i = 10;

do
{
    printf("Hello\n");
}
while (i < 5);
```





Output:
Hello



But the condition is FALSE! Why did Hello print?



Because the order is different!

while:
condition → body
(Check first, then do)



That's the BIG difference!

do-while:
body → condition
(Do first, then check)



Remember:
do-while executes its body
AT LEAST ONCE!

The `do-while` Loop: Syntax & Flow

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Syntax

```
do
{
    statement1;
    statement2;
    ...
    statementN;
}
while (expression);
```

Key point:
The condition is checked after executing the loop body.

Flow of Control

1. Execute
Execute the loop body

↓

2. Check
Evaluate `expression`

↓

TRUE ↓ Repeat

FALSE ↓ Exit loop

From while to for loop!!

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Look at our `while` loop

```
int i = 1;
while (i <= 10)
{
   printf("%d\n", i);
   i++;
}
```

Where are the 3 pieces?

- Initialization: `int i = 1;`
- Condition: `i <= 10`
- Update: `i++`

Notice

They are physically separated.

C gives us a way to



Put the 3 pieces together!

The `for` loop

```
for (i = 1; i <= 10; i++)
{
   printf("%d\n", i);
}
```

Initialization
$i = 1$

Condition
$i \leq 10$

Update
$i++$

`for` Statement – Syntax

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Basic Syntax

```
for (expression1; expression2; expression3)
```

- Expressions can be omitted.
- The semicolons must remain.
- Curly braces can be omitted when the loop contains a single statement.

Most Common Pattern

```
for (initialization; condition; update)
```

Expression 1	Expression 2	Expression 3
Assignment	Relational expression	Assignment
`i = 1`	`i <= 10`	`i++`

Act Out the `for` Loop

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The Code

```
for (i = 1; i <= 3; i++)
{
   printf("%d\n", i);
}
```

$i = 1$

$\downarrow$

$i \leq 3$?

YES $\downarrow$ `printf` $\downarrow$ $i++$ $\downarrow$ CHECK AGAIN

NO $\downarrow$ **EXIT**

What actually happens?

Step-by-step

```
i = 1↓
check:  1 <= 3 ↓
   print 1 ↓
   i++ ↓
check:  2 <= 3↓
   print 2 ↓
   i++ ↓
check:  3 <= 3 ↓
   print 3 ↓
   i++ ↓
check:  4 <= 3↓
```

EXIT

A Tricky `while` Example

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Consider this:

```
int i = 0, j = 0;
while (i < 5, j < 10)
{
    i++;
    j++;
}
printf("%d %d", i, j);
```

Is the condition checking
BOTH
`i < 5` **AND** `j < 10`?

A Tricky while Example

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Consider this:

```
int i = 0, j = 0;
while (i < 5, j < 10)
{
    i++;
    j++;
}
printf("%d %d", i, j);
```

Is the condition checking BOTH `i < 5` AND `j < 10`?

Loop continues while `j < 10`

What is really happening?

`i < 5 , j < 10`

`i < 5` ⟶ evaluated

↓

`j < 10` ⟶ evaluated

↓

value of the LAST expression

Key Point

The **comma operator** evaluates expressions from left to right, but the whole expression gets the value of the **last expression**.

A Tricky `while` Example

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$$(0,0) \rightarrow (1,1) \rightarrow (2,2) \rightarrow \ldots \rightarrow (9,9) \rightarrow (10,10)$$

$$10 < 10 \longrightarrow \textbf{FALSE} \longrightarrow \textbf{EXIT}$$

Output

```
10 10
```

The Loop Trap!!

Consider this:

```
short i;
for (i = 1; i > 0; i++)
   printf("%d\n", i);
```

Output?

The Loop Trap!!

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Consider this:

```
short i;
for (i = 1; i > 0; i++)
   printf("%d\n", i);
```

Output?

Will the loop continue forever?

Max-value $= 2^{15} - 1 = 32767$

Think about the range of `short` int

$$1 \rightarrow 2 \rightarrow 3 \rightarrow \cdots \rightarrow \text{MAX}$$

`i++`

$\downarrow$

What happens beyond the largest represen

Important!

A short integer cannot represent every possible result of an increment.

The C Rule

Incrementing a short integer beyond its representable range causes undefined behavior.

The Infinite for Loop

Consider this:

```
for (;;)
{
   printf("Hello\n");
}
```

OUTPUT?

The Infinite `for` Loop

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Consider this:

```
for (;;)
{
   printf("Hello\n");
}
```

OUTPUT?

- Initialization?
- Condition?
- Update?

Look carefully

```
for (   ;   ;   )
```

Part	Status
Initialization	Omitted
Condition	Omitted
Update	Omitted

⇓

What does that mean?

With no condition to become `false`, the loop keeps executing.

Control-Flow Puzzle: How Many Times?

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code

```
int i = 0, j = 0;
for (i = 0; i < 5; i++)
{
   for (j = 0; j < 1;)
   {
      break;
   }
   printf("Bye\n");
}
```

Predict the Output!

How many times is

Bye

printed?

Control-Flow Puzzle: How Many Times?

॥ सा विद्या या विमुक्तये ॥
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code

```
int i = 0, j = 0;
for (i = 0; i < 5; i++)
{
   for (j = 0; j < 1;)
   {
      break;
   }
   printf("Bye\n");
}
```

Predict the Output!

How many times is

Bye

printed?

Follow the control flow

`i = 0` ↓

`i < 5 ?` ↓

`j = 0` ↓

`break` ↓

`printf("Bye")` ↓

`i++` ↓

Repeat

`break` **exits only the innermost loop.** It does **not** terminate the outer `for` loop.

Solution: Trace the Control Flow

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Step 1: First iteration

$i = 0 \downarrow$

Enter inner loop $\downarrow$

$j = 0 \downarrow$

$j < 1 \quad$ TRUE $\downarrow$

`break` $\downarrow$

$\boxed{\text{EXIT INNER LOOP}} \downarrow$

`printf("Bye\n")` $\downarrow$

$i++$

Solution: Trace the Control Flow

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Step 1: First iteration

$i = 0 \downarrow$

Enter inner loop $\downarrow$

$j = 0 \downarrow$

$j < 1 \quad$ TRUE $\downarrow$

`break` $\downarrow$

EXIT INNER LOOP $\downarrow$

`printf("Bye\n")` $\downarrow$

$i++$

Step 2: Continue the outer loop

Value of `i`	Action
`i = 1`	Print Bye
`i = 2`	Print Bye
`i = 3`	Print Bye
`i = 4`	Print Bye
`i = 5`	**Stop**

`Bye` **is printed 5 times.**

Tricky Example!

Consider this code

```
double k = 0;
for (k = 0.0; k < 3.0; k++);
printf("%lf", k);
```

Question

What will be printed?

Tricky Example!

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Consider this code

```
double k = 0;
for (k = 0.0; k < 3.0; k++);
printf("%lf", k);
```

The ; is the loop body

The loop has an **empty statement** as its body.

Question

What will be printed?

$$k = 0.0 \downarrow$$
$$k < 3.0 \downarrow$$
$$\text{empty body} \downarrow$$
$$k++ \downarrow$$
$$\text{repeat}$$

$$0.0 \rightarrow 1.0 \rightarrow 2.0 \rightarrow 3.0$$
$$3.0 < 3.0 \rightarrow \textbf{FALSE}$$

The `printf` runs only after the loop finishes.
Output:3.000000

Control-Flow Puzzle: `for (5; 2; 2)`

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The Question

```
for (5; 2; 2)
   printf("Hello\n");
```

Output?

Control-Flow Puzzle: `for (5; 2; 2)`

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The Question

```
for (5; 2; 2)
   printf("Hello\n");
```

Output?

Expression 3: 2

2 is evaluated after each iteration, but it does not change the condition.

Expression 1: 5

$\boxed{5} \downarrow$

Evaluated, but its value is **discarded**.

Expression 2: 2

$\boxed{2} \downarrow$

Is 2 zero?

$\boxed{\textbf{NO}} \downarrow$

A non-zero value is **TRUE**.

Therefore

2 is always **TRUE**

The condition is always TRUE. so `Hello` keeps printing indefinitely.

Challenge: Can You Print This Diamond?

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Write a C program using for loop to print:

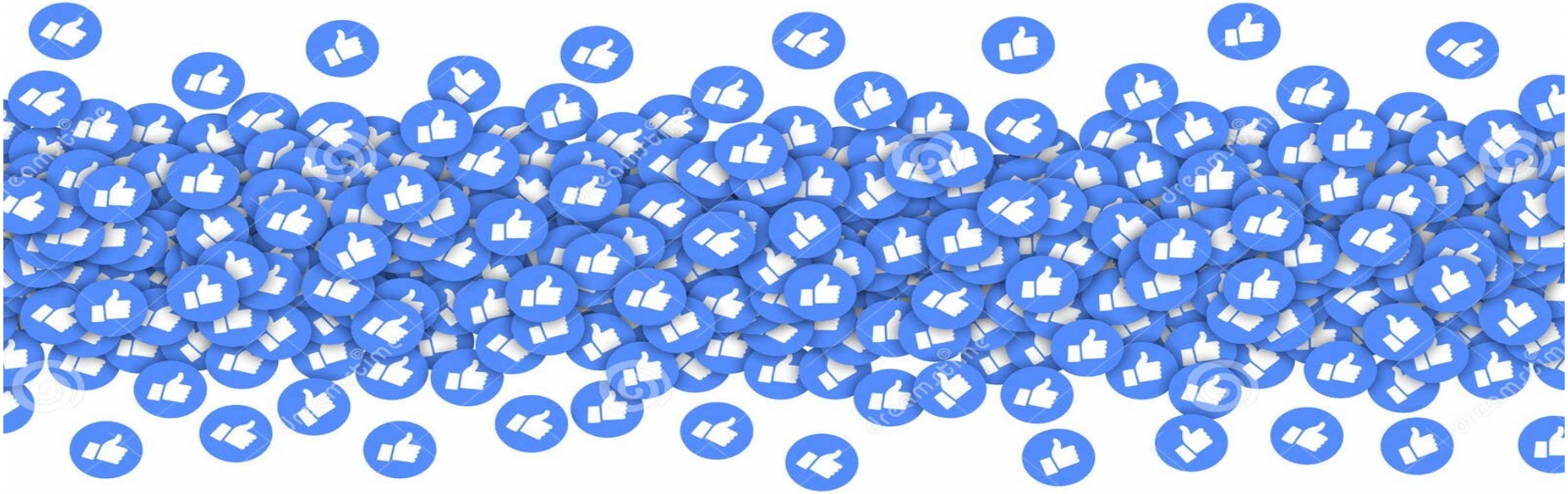
```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

Hint:

Think about spaces + stars + rows

Rule

Use **nested loops**. No hard-coded rows!



Thank you for your kind attention!!